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School or Student? A Mixed Method Analysis on Reddit Data for Transfer Barrier Identification

Phuong Minh Nguyen
Texas A&M University-Victoria
Email: nguyenpminh1608@gmail.com

Donnaban Maldonado
SUNY at Old Westbury
Email: dmaldo15@oldwestbury.edu

Cheyenne Ty
Cal Poly Humboldt
Email: cjt101@humboldt.edu

Yun Wan
Texas A&M University-Victoria
Email: wany@uhv.edu

Xiwei Wang
Northeastern Illinois University
Email: xwang9@neiu.edu

Sherrene Bogle
Cal Poly Humboldt
Email: sherrene.bogle@humboldt.edu

Palvi Aggarwal
University of Texas El Paso
Email: pagggarwal@utep.edu

Shebuti Rayana
SUNY at Old Westbury
Email: rayanas@oldwestbury.edu

ABSTRACT

This study examines the issues that college transfer students experience using Reddit posts. We utilized BERT embeddings, UMAP, and HDBSCAN to cluster topics, and VADER sentiment analysis and manual review to identify two categories of issues: student-centered (e.g., mental health, identity-based exclusion) and school-centered (e.g., credit transfer, advising). This "School vs. Student" viewpoint introduces a new approach to evaluate advising and policy equality. To assess dependability, we used the same procedure to analyze survey data from 161 computing transfer students. Reddit data provided more in-depth and nuanced insights than the surveys.

KEYWORDS: transfer students, Reddit data, BERT embeddings, UMAP, HDBSCAN, sentiment analysis, educational equity

INTRODUCTION

Moving from one community college or university to another may be a difficult and emotionally draining experience, particularly when institutional structures are unable to adapt to students' varied situations. According to previous studies, structural constraints such as confusing financial aid policies, poor advising support, and misunderstandings surrounding credit transfers have been repeatedly observed (Wang et al., 2023; Townsend & Wilson, 2006). However, because these studies mostly rely on surveys and interviews, they might not capture the complex, student-driven viewpoints that influence transfer decisions in the real world.

We examine Reddit conversations as a rich, natural source of student experience to close this gap. Reddit is a useful tool for revealing uncensored issues because of its semi-anonymous style, which promotes open communication. To contextualize our findings, we applied the same analysis pipeline to survey data from MacDonald et al. (2023). While this confirmed many of the same challenges, such as concerns around cost, credit transfer, and advising, the Reddit data revealed deeper and more personal accounts, particularly from students navigating identity-based exclusion, mental health, and marginalization. Using a combination of manual thematic review and contemporary natural language processing (NLP) techniques, such as Bidirectional Encoder Representations from Transformers (BERT) embeddings, dimensionality reduction with Uniform Manifold Approximation and Projection (UMAP), and clustering via Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN), we create a classification framework that separates student-centered issues (personal, emotional, or circumstantial) from school-centered issues (systemic or policy-driven).

In order to better understand the causes of transfer issues, this study presents a "School vs. Student" lens that is generated from two carefully selected subreddits. Text cleaning, sentiment scoring with Valence Aware Dictionary and Sentiment Reasoner (VADER), semantic embedding with BERT, unsupervised clustering, and human-coded thematic validation are all components of our pipeline. The end product is a collection of 16 practical themes, eight for each domain, that can direct future transfer equity research, advisory procedures, and institutional reform.

The main contributions are the following:

Technical Contributions

1. Our mixed-method NLP pipeline extracts meaningful patterns from unstructured student narratives by scraping Reddit using the PullPush API, text preprocessing, Sentence-BERT semantic embedding, UMAP dimensionality reduction, and HDBSCAN clustering.
2. We utilized VADER sentiment analysis to identify emotionally charged situations and inform theme coding, thereby gaining a more detailed understanding of student pain points and frustrations.
3. After recurrent manual review, we developed a dual-domain thematic framework with 16 categories (eight student-centered and eight school-centered) based on semantic similarity, sentiment patterns, and student expressiveness.
4. Our study found that Reddit provides access to underrepresented or emotionally sensitive viewpoints, particularly on identity, mental health, and institutional trauma, that are not often recorded in survey research.

Practical Contributions

1. Our methodology helps college counselors, student affairs staff, and policymakers identify and address both institutional (school-centered) and personal (student-centered) barriers to successful transfer.
2. These themes guide the creation of AI-powered advising systems by recognizing non-academic factors, including emotional preparedness, identity affirmation, and advising clarity, that are crucial for individualized suggestions.
3. Social media data can enhance traditional survey techniques by revealing hidden aspects of student experience, leading to more effective intervention design.
4. This study establishes the framework for a Transfer Equity Index, a data-driven tool that institutions may utilize to measure and improve their response to varied transfer student requirements.

LITERATURE REVIEW

Transfer students are a growing and important part of the higher education landscape. Nearly 38% of college students in the United States transfer at least once every six years (Shapiro et al., 2018). These students have various academic backgrounds and life experiences, but they also suffer significant transitional problems that are frequently underestimated.

Historically, research on transfer students has focused on two stages: academic performance after admission (e.g., GPA maintenance, course engagement) and post-transfer outcomes like persistence and degree completion. Previous research using the Social Cognitive Theory and STEM Transfer Models, for example, has linked self-efficacy, outcome expectations, and institutional receptiveness to students' long-term success after transferring, especially among computing and underrepresented minority students (Ty et al., 2024; Vargas et al., 2022). These discoveries have influenced support programs designed to assist transfer students in adjusting and thriving once they have enrolled at a new university.

However, few studies have centered on the early transfer process, the moment at which students grapple with ambiguous policies, social isolation, emotional hurdles, and limited support. This early-stage gap is crucial because the problems experienced throughout the transfer process frequently impact students' long-term trajectories and well-being.

Prior research on college transfer has generally focused on finding structural and demographic indicators of student success, with a focus on GPA, credit accumulation, and graduation rates (Blekic et al., 2020; Ishitani and Flood, 2018). Much of this study is based on institutional statistics, standardized surveys, or interviews, with an emphasis on measurable outcomes rather than the complex reality of student voice, which may overlook the more truthful, emotional, or identity-centered experiences that influence transfer decision-making. Townsend and Wilson (2006) and Maliszewski and Hayes (2020) found gaps in advising and emphasized the need of transfer student capital (TSC), including knowledge, support networks, and self-advocacy. These data show that under-resourced advising systems frequently force students to rely on informal help from classmates and family during transitional periods.

Social media has recently emerged as a supplementary lens in literature. For example, Melton et al. (2021) and Yan and Liu (2022) used Reddit to record public standpoints and conversations on health and education themes. Standfast et al. (2024) expanded this research to transfer

students, revealing student concerns on a large scale, utilizing topic modeling and sentiment analysis. Their investigation, however, focused largely on issue frequency and emotion polarity, rather than delving deeper into the nature of the impediments that students experience. Furthermore, most existing research conflates institutional and personal obstacles, making it difficult to identify systemic vs individual explanations of transfer friction.

To replenish this gap, our study examines unstructured Reddit postings to get real-time, student-driven experiences of the transfer process. We go beyond standard success indicators by establishing a categorization approach that distinguishes between school-centered (policy, advising, systems) and student-centered (mental health, identity-based exclusion, caregiving) concerns to uncover underlying elements that influence transfer equality and experiences. By using BERT embeddings, UMAP dimensionality reduction, and HDBSCAN clustering to Reddit postings, we can uncover not just what students are talking about, but also why these worries arise, whether they are founded in institutional design or personal experience. Compared to previous survey-based research, the adoption of Reddit allows for more in-depth access to underrepresented, emotionally sensitive narratives, notably those of neurodivergent and LGBTQ+ students whose experiences may not be captured in regular institutional databases. This categorization serves as the framework for more focused advising initiatives, as well as a future Transfer Equity Index that can measure institutional responsiveness across both domains.

MOTIVATION

There are several different types of transfer students. Some people find it relatively easy to manage the procedure, but many others encounter significant obstacles, such as identity-based exclusion, mental health issues, and unclear credit transfer laws. Forms, checklists, and deadlines are examples of procedural compliance that are regularly emphasized in institutional replies, while students commonly look to unofficial sites like Reddit for both practical guidance and emotional support.

An important point is brought up by this disconnect: Do institutional structures fail to take into account the varied realities of transfer students, or are these difficulties mostly the result of student misunderstanding? Our study examines the ways in which students express their experiences in public settings in order to tackle this issue.

Using sentiment analysis and clustering to examine Reddit discussions, we want to:

- Determine the underlying cause of transfer difficulties (personal vs. institutional).
- Capture themes that are frequently missed in institutional data or standard surveys.
- Offer a diagnostic perspective to enhance the mechanisms for student support and advising.
- Establish the framework for a Transfer Equity Index that is driven by data.

Policy, advising practices, and equity audits can all benefit from the insights this work produces, which will help institutions better differentiate between issues that call for systemic change and those that require focused student assistance.

METHODS

This work extracted and analyzed Reddit posts about the experience of transferring to college

using a mixed-methods NLP pipeline (see Figure 1). We integrated text preprocessing, sentiment analysis, clustering, semantic embedding, and manual theme evaluation into our multi-phase approach. The fundamental objective was to divide emerging topics into two categories: student-centered and school-centered.

Figure 1: Methodology Workflow

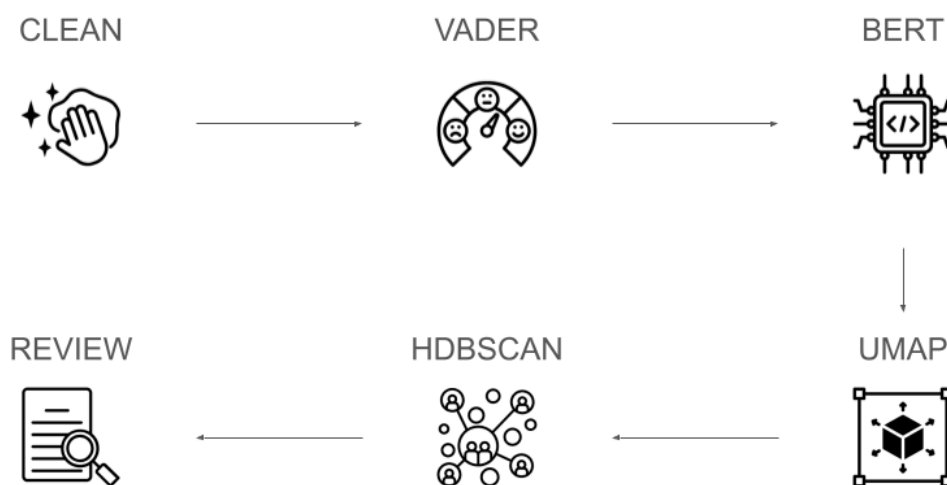


Figure 1 illustrates the methodology workflow used in the NLP pipeline. The process begins with data cleaning, followed by sentiment analysis using VADER. Text embeddings are then generated using a BERT model. These embeddings are reduced in dimensionality with UMAP and clustered during HDBSCAN. Finally, the resulting clusters are manually reviewed for thematic relevance.

Data Collection

We gathered Reddit posts using the PullPush API (formerly known as Pushshift), a well-known technique for large-scale data extraction in academic research (Baumgartner et al., 2020). Two subreddits, r/TransferStudents and r/CollegeTransfer, were chosen because they have active conversations about transfer topics. These subreddits were also chosen due to their relatively large size and generalizability to any college.

After the extraction of approximately 27,000 posts, they were cleaned and inspected to ensure they were relevant to the college transfer experience. Relevance was judged by whether a post included personal transfer experiences, institutional regulations such as credit transfer or GPA requirements, emotional or logistical difficulties such as stress or misunderstanding, or worries about social fit, identity, or campus life.

We used rule-based filters to remove entries that were too short (less than 25 words), irrelevant to transfer (such as graduate school admissions or general housing questions), duplicates, possibly spam, or auto-generated. Posts that included no apparent evidence of transfer-related information were also deleted.

Approximately 1,000 high-quality posts were saved and prepared for sentiment and theme analysis.

Sentiment Analysis

To further understand the emotional dynamics of transfer students' experiences, we added sentiment analysis to theme clustering as a supplemental layer. Sentiment reveals how students feel about the subjects they discuss, whether they show dissatisfaction, optimism, bewilderment, or satisfaction. By categorizing posts as good, negative, or neutral, we can find emotional patterns that are associated with various transfer-related hurdles or supports. Negative mood, for example, frequently indicates pain whether it comes to advising, credit evaluation, or institutional communication. Positive emotion, on the other hand, might emphasize successful techniques or helpful circumstances. A neutral attitude often represents factual or informative content, which helps to contextualize the analysis.

We used the VADER (Valence Aware Dictionary for Sentiment Reasoning) sentiment analyzer because of its excellent performance on social media-style content, which frequently contains informal syntax, slang, and expressive punctuation. VADER combines a lexical dictionary with heuristics to handle intensifiers, negations, and punctuation, all of which are frequent in Reddit conversations. Each post received a compound emotion score that ranged from -1 (very negative) to +1 (highly positive). Following VADER's preset criteria, posts with a score of less than -0.05 were classified as negative. These posts were selected for a more in-depth thematic evaluation because they frequently express emotional suffering or perceived inequality.

However, because VADER is a rule-based system, it may underperform on messages that are extremely contextual, caustic, or unclear. To address this, sentiment ratings were viewed as indicators for more manual assessment rather than definitive judgments of emotional tone.

Text Embedding

After cleaning the text, we encoded each Reddit post using all-MiniLM-L6-v2 (Reimers & Gurevych, 2021), a pretrained Sentence-BERT model designed for semantic similarity and clustering applications. This transformer-based methodology produces condensed, contextualized vector embeddings that contain sentence-level meaning. This method, as opposed to classic models such as TF-IDF or GloVe, allows for more accurate semantic comparisons of posts that use diverse language to communicate comparable problems. In the next step, these embeddings were used to reduce dimensionality and cluster data.

Dimensionality Reduction & Clustering

To minimize the dimensionality of phrase embeddings while keeping their semantic structure, we used Uniform Manifold Approximation and Projection (UMAP; McInnes, Healy, & Melville, 2018) with $n_neighbors=15$, $n_components=5$, $min_dist=0.0$, and a cosine similarity metric. This setup enables us to preserve fine-grained semantic links between posts in a lower-dimensional space.

We next grouped the shortened vectors with HDBSCAN, a density-based clustering technique that works well with noisy and variable-length social media texts. In the reduced space, we

employed Euclidean distance with a minimum cluster size of 5. This arrangement allowed for the finding of compact, high-resolution topic clusters without predefining the number of groups. The parameters were empirically calibrated to strike a compromise between coherence, cluster granularity, and noise suppression.

Figure 2: Theme Distribution

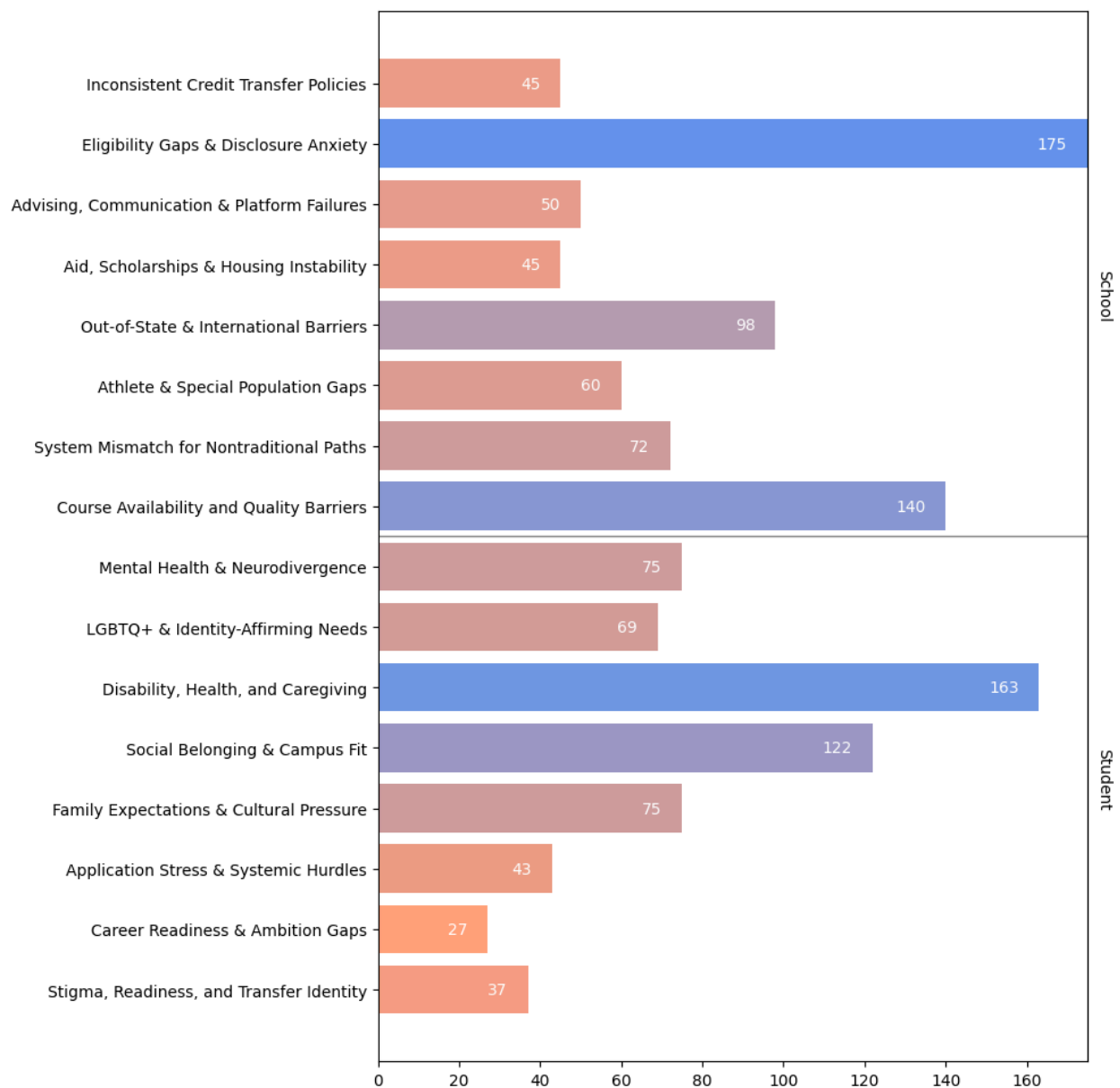


Figure 2 presents the distribution of themes across the two identified domains: student-centered and school-centered issues. Notably, eligibility gaps and disclosure anxiety are the most prominent school-centered concerns, while disability, health, and caregiving emerge as the most prominent student-centered challenges.

Manual Theme Review

Each cluster was thoroughly reviewed and labeled using the top keywords, representative stories, and related sentiment scores. The labeling approach used an iterative coding procedure, with initial codes created after a preliminary evaluation of representative conversations from each cluster. These initial codes were then validated and improved through numerous rounds of cluster comparison to guarantee consistency, coherence, and theme saturation. Discrepancies or overlaps were handled by re-evaluation and discussion, which resulted in changes to definitions or the merger or separation of categories. This iterative approach persisted until consistent and clear motifs emerged. Finally, 16 theme categories were chosen, eight of which were school-centered and eight of which were student-centered, based on repeating language and emotional patterns found in the dataset.

Final Classification Framework

One of two domains was assigned to each theme in the final classification:

- **School-Centered:** Obstacles brought on by institutional structures, rules, or communication failures (e.g., unclear credit transfer restrictions, aid delays).
- **Student-Centered:** Personal context-based difficulties, such as those related to identity, mental health, caregiving duties, or social belonging.

RESULTS

Overview

Using two carefully selected subreddits, we found and categorized sixteen theme groups. Either the School-Centered or the Student-Centered overarching domains were assigned to each post. The final framework, which was produced via unsupervised clustering and manual validation, demonstrated how transfer students express their obstacles and what causes the greatest anxiety on the internet. It produced 16 fundamental issue types, 8 for each area.

Final Thematic Framework

School-Centered Issues

Inconsistent Credit Transfer Policies. Transfer students sometimes experience broad ambiguity as a result of irregular credit transfer regulations that differ by school. The acceptance of Advanced Placement (AP) and dual enrollment credits is variable, leaving students wondering how their previous work will be counted. Complicating matters, paths such as IGETC and AS-T specific to California community colleges are inadequately articulated, making academic preparation difficult and unreliable. Transfer times are rarely transparent, resulting in delays and weakening students' trust in the process. Dual enrollees are occasionally misclassified during the application review process, and overseas transcripts are routinely delayed owing to ambiguous evaluation standards. Overall, these systemic inefficiencies lead to academic failures such as GPA loss, course retakes, and crushing frustration - problems that transfer students must manage on their own with inadequate institutional help.

Eligibility Gaps & Disclosure Anxiety. In addition to these logistical hurdles, students experience anxiety as a result of confusing qualifying standards and opaque institutional procedures. Poorly

stated policies, such as the 90-unit cap, are a major cause of worry for many students, who hear about them only after they have an impact on their transfer plans. Inconsistent advice exacerbates the problem, generating mistrust and uncertainty when multiple consultants provide contradicting advice. Honest errors in applications (often due to a lack of transparency) are punished unfairly, making students feel penalized for attempting to navigate a complex system. International students are particularly vulnerable since they frequently get insufficient information concerning eligibility, resulting in even greater anxiety. The inclusion of ambiguous regulations turns what should be a simple compliance procedure into an emotional burden. Underlying all of this is a rising worry that schools value status above student success, inhibiting open disclosure and exacerbating the dread of being rejected on technicalities.

Advising, Communication & Platform Failures. A widespread failure in communication and advisory channels adds to the anxiety caused by policy opacity. Students report not obtaining a response after multiple outreach attempts, leaving them trapped at important decision moments. Even when communication does occur, competing advice between departments undermines confidence and causes more confusion than clarity. Students are sometimes dismissed with comments like "check the website," which sound impersonal and useless, especially when the websites themselves are out of current or lack the necessary information. Technical failures worsen the situation: platform malfunctions and app crashes result in lost effort, such as retyping entire essays, which can cause aggravating delays. The whole system feels mechanistic rather than human-centered, leading students to look for community-driven alternatives. As a result, many people use Reddit and other peer forums not for informal conversation, but as a last option when institutional advice fails.

Aid, Scholarship & Housing Instability. When institutional guidance and digital systems fail, students' access to fundamental needs, most notably financial aid and housing, suffers as a result. Scholarship criteria are frequently ambiguous, prompting students to lose funds unexpectedly. Financial aid consequences for changing course loads are rarely disclosed, generating a sense of insecurity while making ordinary academic decisions. Delays in the disbursement of financial prizes can directly prevent access to university housing, particularly for individuals with limited finances. Transfer and foreign students are disproportionately disadvantaged, as they are structurally excluded from many aid programs and get little specialized assistance. Furthermore, there is a clear gap between aid programs and student well-being; regulations never take into account the lived realities of stress, debt, and housing instability. Many scholarships also require full-time attendance, which puts additional pressure on students who may require part-time flexibility to balance employment, health, and caregiving commitments. In the absence of timely assistance, these financial impediments become disruptive influences in students' academic lives.

Out-of-State & International Barriers. The financial assistance issues that transfer students face are exacerbated for those who are out-of-state or overseas, since geographic status adds another layer of exclusion. Strict residence requirements sometimes devalue nonresident applications, creating the impression that quality takes a second place to zip codes. International students confront much greater challenges since they are ineligible for FAFSA aid and are sometimes excluded from institutional support networks entirely. Across all categories, there is a noticeable dearth of advice targeted to their needs, forcing students to tackle complicated tasks alone. Even highly qualified applicants report being waitlisted without explanation, confirming the impression that they are being rejected because of their background rather than their academic ability. When policies remain mute on these injustices, students see it

as an institutional message that they are not welcome. As a result, there is a continual undertone of anxiety and discouragement, even among high-achieving out-of-state applicants, who believe that regardless of their performance, their chances are already bleak.

Athlete & Special Population Gaps. Aside from geographic location, institutional systems fall short in servicing scholars from specific demographics, notably athletes, students with disabilities, and those whose requirements do not fit the standard model. Academic assistance for athletes frequently decreases upon transfer, especially as scholarship conditions become unclear or disturbed. Intensive training schedules might interfere with course availability, delaying academic achievement. Meanwhile, students with disabilities confront a different but equally aggravating challenge: they must continuously resubmit papers to keep the accommodations they were previously authorized for. In many situations, institutional assistance appears to be centered on prescribed roles, such as athlete, international student, or student with a handicap, rather than a holistic view of the student as a whole. As a result, schools often serve administrative categories rather than real human experiences. These students may abruptly lose access to the services, resources, or accommodations on which they had previously relied, compounding a sense of isolation and institutional indifference.

System Mismatch for Nontraditional Paths. Students on atypical academic routes face a system that was never designed with their specific needs in mind. Military duties, professional commitments, and caregiving responsibilities frequently clash with strict academic schedules, and students who take gap years or follow varied timetables are penalized for straying from the norm. As institutions continue to function under the presumption of conventional, full-time enrollment, online and part-time students are routinely disregarded in policy formulation and advising. This structural bias also applies to individuals who are older than traditional college-age students (typically 18–22 years old), who have less flexibility while balancing complex living situations. Application forms and institutional regulations frequently lump applicants into simple groups, neglecting the intricacies of diverse educational, professional, and personal experiences. As a result, people who do not fit a restricted concept of the "standard student" are pushed to shape their lives to fit the system, rather than the system adjusting to meet the different requirements of actual students.

Course Availability and Quality Barriers. To round out the school-centered difficulties, course availability and quality are important hurdles that disrupt the transfer pathway, not just academic accomplishment. Transfer students commonly discover that important prerequisite courses necessary for articulation are missing, discontinued, or offered rarely, impeding their ability to satisfy transfer eligibility. Scheduling problems exacerbate the problem, especially when compulsory courses interfere with students' employment, caregiving responsibilities, or transportation constraints. Multi-semester gaps between courses impede transfer times in disciplines with rigorous sequencing requirements, such as STEM. Aside from access, the quality and alignment of courses can cause friction: community college programs may lack the depth or requirements of receiving schools, causing students to repeat material or arrive underprepared. Inconsistent articulation agreements and imprecise preparation criteria result in credit loss and disrupted academic objectives. Finally, these institutional inadequacies in course availability and difficulty do more than merely delay students; they actively disrupt transfer ambitions.

Figure 3: Final Thematic Framework

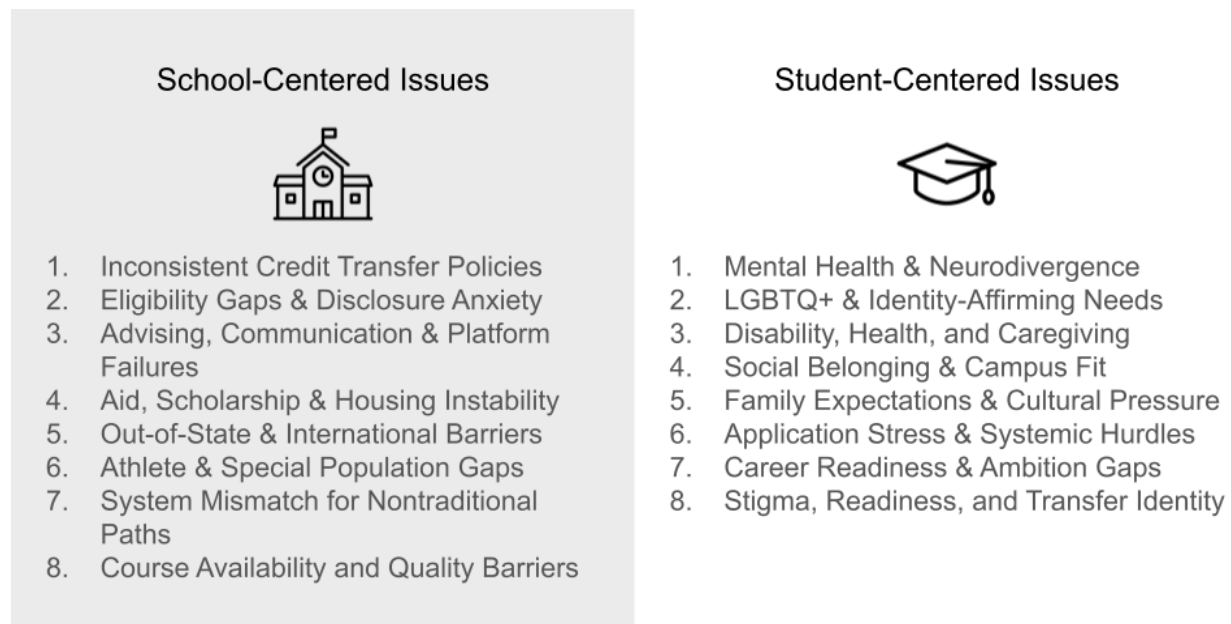


Figure 3 presents the final thematic framework, categorizing transfer-related issues into school-centered and student-centered domains. School-centered themes highlight institutional hurdles, such as policy uncertainty or platform failures, whereas student-centered themes focus on identity, mental health, and personal context. This structure is the basis for the Transfer Equity Index.

Student-Centered Issues

Mental Health & Neurodivergence. While institutional flaws generate structural hurdles, transfer students also face profoundly personal issues, none more serious than mental health and neurodivergence. For many people, mental health issues had a direct impact on their decision to transfer, frequently in reaction to circumstances that lacked psychological safety and flexibility. Taking medical leave exacerbated anxieties of being judged or disqualified, and burnout was commonly faced in secret due to shame. Students indicated a need for actual, empathetic assistance, not simply policies or booklets. When it came to picking a new school, inclusion and care took precedence, indicating that the transfer process is as emotional as it is scholastic. Despite this, admissions teams frequently overlooked emotional requirements in favor of transcripts and test scores. These data show that transferring is more than simply a logistical step for many students; it is also a survival strategy.

LGBTQ+ & Identity-Affirming Needs. Many transfer students, particularly those who identify as LGBTQ+, underlined the importance of identity-affirming environments, which are closely related to mental health. For these students, safety and identification were not secondary considerations; they were crucial to the transfer choice. Experiences of being erased, misgendered, or unsafe on campus caused substantial emotional suffering, particularly in dorms and classrooms. Students actively sought exposure, belonging, and inclusion, rather than simply tolerating others. Many people picked organizations that stressed care above

prominence, indicating a deeper desire for environments that validated who they were. Judgment was frequently feared for components of identity, such as gender expression or sexual orientation, rather than academic performance. For some, this load was exacerbated by intersecting responsibilities, such as being LGBTQ+ and a caregiver, making support systems even more important. These data confirm that transfer decisions are frequently motivated by the desire for dignity rather than degrees.

Disability, Health, and Caregiving. Many transfer students had to deal with chronic health concerns, impairments, or caregiving duties, which hampered academic progress and influenced their educational choices. Students coping with long-term sickness frequently had enrollment gaps, while caregivers struggled to locate institutions that provided the flexibility required to meet their combined obligations. Even when accommodations had previously been accepted, they frequently did not transfer smoothly, forcing students to repeatedly justify their needs - an extremely demanding procedure. Some students, particularly those with religious or sensory requirements, felt alienated from institutions that valued conformity above response. Environmental issues, such as seasonal sadness or weather-related flare-ups, made things more difficult, especially when schools failed to provide appropriate scheduling or learning modes. What students plainly articulated was the need for systems that adapt to human complexity rather than inflexible frameworks that ignore it. Many people choose to transfer to find organizations that accept, rather than oppose, the reality of their existence.

Social Belonging & Campus Fit. Even when academics were solid, many students decided to transfer because they did not have a sense of social belonging at their former university. Disconnection was prevalent, even at highly rated or prominent schools, since students missed the community and support networks they had established elsewhere. Large campuses can feel isolated and impersonal, especially for older or returning students who struggle to locate peer groups with whom they can identify. In addition to academic offers, students sought deeper alignment and schools where they felt intellectually and socially at ease. Belonging was more than simply clubs or orientation programs; it was about feeling noticed, supported, and absorbed into a common academic and social culture. Many students deliberately sought out research-oriented, aspirational societies that fit their desire and interest, rather than merely their transcript. For these students, moving meant more than just getting out of a difficult situation; it meant looking for a place where they could succeed.

Family Expectations & Cultural Pressure. Many students from marginalized backgrounds, particularly first-generation, Latinx, and immigrant families, made transfer decisions for reasons other than academics. They were influenced by strong cultural norms and emotional duties. For these students, moving typically meant facing disappointment or censure from family members who valued stability and conventional pathways. Some people stayed in subpar academic situations to avoid feeling like they failed or were ashamed. Others relocated closer to home to alleviate financial stress or fulfill caregiving responsibilities, and pressures that disproportionately impact women and vulnerable communities. Each transfer tale had underlying themes of obligation, remorse, and sacrifice, illustrating the ongoing struggle between personal development and communal responsibility.

Application Stress & Systemic Hurdles. For many transfer students, the application process has become a cause of great stress and uncertainty. The stairs were sometimes tiring, particularly for those taking unusual routes. Online and military students faced distinct obstacles, since many lacked access to recommenders or did not know instructors well enough to acquire letters

of recommendation at all. Systems and forms felt out of date and rigid, built on assumptions that did not represent students' daily circumstances. Applicants typically viewed the procedure as "built for someone else," with inflexible designs that excluded online education, interrupted timetables, or hybrid learning. What should have been a portal to opportunity felt more like a barrier that punished students for taking different educational paths. For many, the most difficult aspect of moving was not the choice but the documentation.

Career Readiness & Ambition Gaps. For many transfer students, the decision to change schools was motivated not just by what was not working but also by what they hoped to achieve, particularly in terms of professional objectives. Career services at their initial schools were frequently viewed as antiquated, impersonal, or lacking in real opportunities. Students reported restricted access to internships, research experience, and professional mentorship, all of which are essential for starting a career. Some switched intentionally to obtain access to more robust alumni networks, ambitious peer groups, and industry-focused programs. They weren't just looking for a job; they were looking for a long-term match with their goals. The requirement for real-world preparation, such as job interview training and business connections, was frequently overlooked, particularly among first-generation or disadvantaged students. In many situations, selecting a new school meant committing to a brighter future.

Stigma, Readiness, and Transfer Identity. Many students' experiences were accompanied by a quieter but profoundly felt burden: the stigma of being a transfer student. Some felt criticized for beginning at a community college or transferring to an institution seen as "lower ranked," which influenced how they were perceived by peers, advisors, and even themselves. This outward and internalized stigma harmed students' self-esteem, motivation, and sense of academic identity. Burnout and mental exhaustion were frequently mistaken for laziness or a lack of drive, rather than as indicators of hard work and persistence. Emotional preparedness was rarely emphasized in advising and application situations; evaluations focused on transcripts rather than the person behind them. Students discovered that their unconventional aspirations frequently conflicted with institutions that expected straight, standard routes. Despite making purposeful and smart transfer decisions, many students questioned their own value. What they desired most of all was affirmation and respect, not pity or reduced expectations.

Sentiment Distribution

Negative Posts

Negative sentiment was found in around 11% of Reddit posts and 8% of survey replies, according to VADER sentiment research. While this may appear to be a modest number given the prevalence of transfer-related issues, it demonstrates VADER's dependence on overt emotional signals. Students frequently expressed institutional impediments such as GPA loss, credit rejection, or inadequate advising, in layman's words, which VADER perceived as neutral or even positive. Because of their overall tone, common comments such as "It was tough, but I'm glad I transferred" are usually reported as positive. The model also struggled to recognize sarcasm, emotional restraint, and suppressed dissatisfaction, underestimating discomfort in posts about severe academic or psychological obstacles. Despite the low sentiment score, many of these narratives focused on hardship and bewilderment, particularly on topics such as scholarship instability, identity struggle, and advisor mistrust. Statements like "I feel punished for being honest" or "I didn't know I'd have to start over" hint at a deeper emotional reality that VADER's rating does not completely capture.

Figure 4: Sentiment Distribution: Reddit vs Survey

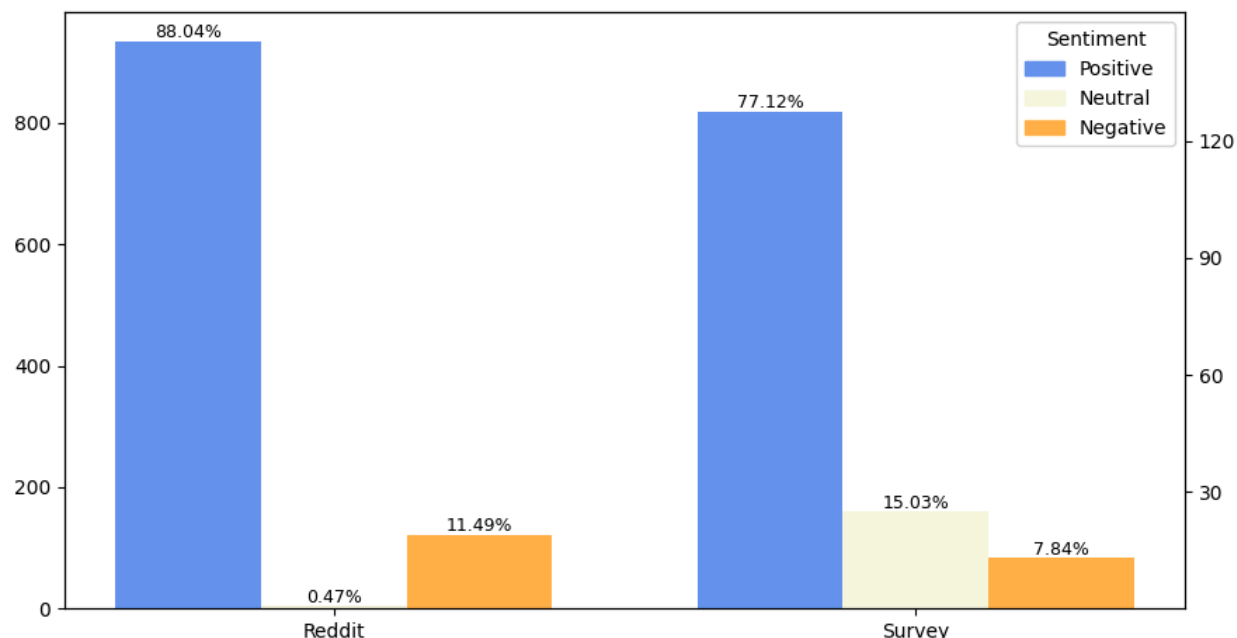


Figure 4 depicts the distribution of sentiment across both the Reddit and survey data. Compared to Reddit, the survey data shows a 10.12% decrease in positive posts, a 3.65% decrease in negative posts, and a 14.56% increase in neutral posts, suggesting that survey responses were more restrained or neutral in tone, potentially due to the structured format or perceived formality of the survey setting.

Positive Sentiment and Resilience

Positive sentiment prevailed in both datasets, occurring in 88% of Reddit posts and 77% of survey replies. This shows that, even when faced with hurdles, many students saw their transfer experience as ultimately beneficial. Optimism was associated with themes of personal development, mental health rehabilitation, and academic regeneration, particularly in Reddit postings. The phrases "the best decision I ever made," "my mental health improved after transferring," and "finally found the right fit" were often used. Survey responses reflected this resiliency, but with greater complexity and neutrality. Many of the participants recognized obstacles while expressing optimism or thanks for the outcomes. This distribution emphasizes transfer students' ability to recast adversity as change, and it implies that emotional tone, while partially captured by methods like VADER, frequently combines suffering with healing.

Survey Comparison

Using the same methodology, we processed survey data from MacDonald et al. (2023). The survey included five open-ended questions for Computer Science students at four-year institutions and two for community college students. The first two questions were answered by both groups, while the remaining three were specific to four-year students. The questions were:

- *Q1: General Information:* Please list the information that you used when deciding between a community college and a 4-year university. Please also mention the information that you feel was needed but not available. (For example 4-year university nearby, better job opportunities at a 4-year university, ease of transfer, cost, guidance from family, friends, etc.)
- *Q2: Important Information:* Please indicate the most important information from the above list that would affect the decision of transfer.
- *Q3: Challenge:* What challenges did you encounter as you were transferring from your previous college or university?
- *Q4: Expectation:* How did your expectations or career plan change after transferring to your current institution?
- *Q5: Experience:* Please share any particular experience you feel is important as a transfer computing major student.

Figure 5: Reddit vs Survey Theme Comparison Chart

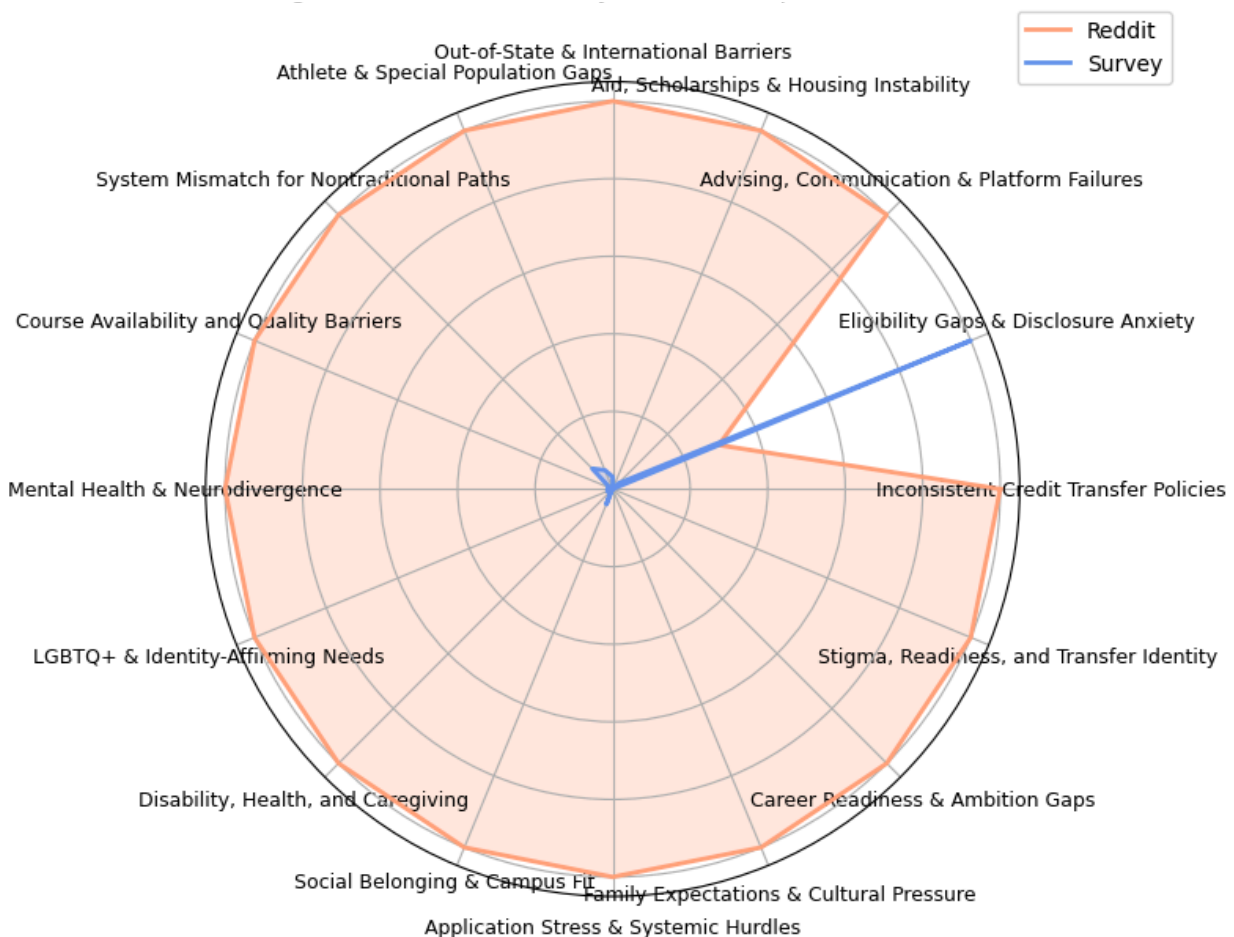


Figure 5 compares the prevalence of themes across Reddit and survey data. While Reddit posts showed broad coverage across all themes, survey responses were more narrowly concentrated, with higher emphasis on experiences within Eligibility Gaps & Disclosure Anxiety, highlighting a

more focused but limited range of concerns in formal survey settings.

Survey Findings

The methodology produced 6 clusters for four-year students and two for two-year students. Upon manual review, the following themes were identified.

1. **Transfer Preparedness and Informal Guidance.**
This cluster reflects a general lack of understanding and preparedness in the transfer process, with students often relying on informal support from family and friends. A frequent use of vague or minimal responses like “n/a” suggests low engagement, limited confidence, or uncertainty about the transfer process.
2. **Career-Oriented Transfer Decisions**
This cluster highlights students who approach the transfer process with clear career goals in mind. Their decisions were driven by the pursuit of better job opportunities, internships, and alignment with long-term career plans. Many viewed community college as a strategic stepping stone, carefully planning their transfer and selecting majors with future outcomes in mind.
3. **Convenience and Practical Considerations**
This cluster centers on practical and logistical factors that influence students’ transfer decisions. Affordability was a primary motivator, but students also emphasized ease of transfer, streamlined processes, and minimal bureaucratic challenges. Considerations like commuting, housing, scheduling flexibility, and program availability played significant roles. The overall tone suggested a pragmatic, solution-focused transfer experience.
4. **Opportunities and Support**
This cluster reflects students driven by opportunities for growth, both academic and personal. Unlike the career-oriented group, their focus extended beyond jobs to include self-improvement and productivity. Support from peers and online communities like Reddit and Discord played a meaningful role. Although some challenges were noted, including racism, health issues, and family circumstances, the overall tone remained positive and future-focused.
5. **Cost and Credit Transfers**
This cluster centers on cost as the dominant factor in transfer decisions, more strongly than in any other group. While cost appeared to some degree across all clusters, students here emphasized affordability as the ultimate influence, often citing it as the reason for starting at a community college. Posts also focused on the mechanics of transfer, such as credit evaluations, prerequisites, and advising.
6. **Advising, Barriers, and Identities**
This cluster highlights the complexity of the transfer student experience, shaped by advising gaps, systemic barriers, and identity-related struggles. Students describe poor or absent advising during key decision points, alongside legal and institutional obstacles such as visa issues or undocumented status. Many reflected on identity-related challenges that contributed to self-doubt and imposter syndrome, such as being first-generation students or part of underrepresented groups.

Comparison

The resulting clusters did not reveal any findings that were entirely new compared to MacDonald et al. (2023), but they did reinforce key concerns identified in the Reddit analysis, particularly around cost, credit transfers, and institutional barriers.

More importantly, using the same methodology for both datasets provided a clear way to evaluate the value of social media data in this type of research. While the survey responses captured important challenges in the transfer experience, they lacked the depth and detail found in Reddit posts. Social media data revealed more specific and often overlooked concerns, including the needs of LGBTQ+ students, neurodivergent individuals, and students with disabilities. Even on topics that appeared in both sources, Reddit posts offered more context and richer accounts than the survey responses. This difference may partly stem from how Reddit fosters more open, candid conversations, whereas the clinical tone of formal surveys may discourage students from sharing vulnerable or stigmatized experiences.

DISCUSSION AND CONCLUSION

Through the classification of Reddit posts into school-centered and student-centered categories, this study presents a novel framework for comprehending college transfer difficulties. This approach records genuine, unsolicited observations from students navigating the transfer process in real time, in contrast to standard survey-based research, which could overlook subtle or developing concerns.

Our research shows that a large number of systemic problems, such as imprecise credit procedures, automated advising systems, and scholarship disqualifications, have institutional roots. Students are frequently left perplexed, unsupported, or misinformed by these school-centered obstacles. Student-centered difficulties, on the other hand, represent identity, health, emotional labor, ambition, and realities that are often overlooked by institutions in their advising frameworks.

Importantly, our comparison between Reddit and survey data revealed a noticeable gap in the depth and range of issues that surfaced. While the survey captured many core challenges, the Reddit posts brought forward more personal and often marginalized experiences. There are two possible explanations for this difference. First, the anonymity of Reddit may make students feel safer discussing sensitive topics such as mental health, identity, or discrimination, issues they might hesitate to disclose in a formal survey. Second, Reddit likely draws from a broader and more diverse population of transfer students, including those whose experiences are not typically represented in academic research. This suggests that social media data can play a critical role in surfacing overlooked perspectives and complementing traditional research methods.

The "School vs. Student" dichotomy has useful applications:

- It points out places where processes might be changed for institutions in order to lessen harm and increase clarity.
- It provides advisers with a diagnostic checklist to help them better grasp the underlying reasons why students experience stress.
- It gives academics a step closer to creating a Transfer Equity Index, a standard instrument for gauging an institution's responsiveness to various transfer requirements.

We also identified high-impact annoyance sites by examining clusters of negative sentiment. Students were perplexed by conflicting aid regulations, felt penalized for taking medical leaves, and were misinformed about GPA standards. Rather than student failure, policy ambiguity is frequently the cause of these situations.

Practical Implications

This study offers valuable data for a variety of critical stakeholders, including college advisers, policymakers, AI system developers, and student assistance programs. The classification of issues into school-centered and student-centered categories provides institutions with a clear mechanism to analyze and improve their transfer-related activities. For example, schools might utilize this approach to identify and fix issues such as confusing credit regulations, broken advice lines, and unreliable financial assistance systems.

Advisors can also utilize the framework as a diagnostic tool to help understand the more complex issues that students confront. These include mental health issues, neurodivergence, caregiving responsibilities, and identity-based stress, which are frequently overlooked in typical advising approaches. By recognizing these elements, advisers may provide more humane and empathic assistance.

The themes revealed in this study have the potential to be essential elements for AI-based advisory systems. Incorporating emotional and contextual aspects into the system design will result in more customized and accurate suggestions, particularly for students who may not pursue standard academic courses.

Finally, the usage of Reddit data demonstrates the utility of social media as a research tool. Many sensitive or marginalized transfer experiences, such as those involving mental health, prejudice, or identity, were more vividly expressed on Reddit than in traditional questionnaires. This shows that while developing policies and support systems, institutions should include real-world student perspectives. Our findings also lend support to the creation of a Transfer Equity Index, a potential statistic for assessing how effectively institutions facilitate different transfer experiences.

LIMITATIONS AND FUTURE WORK

Limitations of this study include the sample size (about 1,000 posts), the bias of the Reddit platform (English-speaking, internet-savvy individuals), and the inability to clearly identify student identities or transfer patterns (e.g., lateral shifts from 2 years to 4 years versus 4 years). We also found that a large portion of the posts were concentrated within the California university system. This is likely due to its large number of students and campuses, though it may introduce some bias. Furthermore, subjectivity was introduced by manual labeling, even if it guaranteed theme accuracy.

In subsequent work, we intend to:

- Extend the system to more extensive and varied Reddit datasets
- Consider incorporating Zero-Shot subject classification to enhance automation
- Examine answer quality (do pupils receive correct guidance?)
- Create a dashboard tool for advisers using this framework
- Create a prototype. Theme frequency and sentiment weight are used in the Transfer Equity Index

Transfer Equity Index

To put our results into practice, we propose the Transfer Equity Index (TEI), which is a composite diagnostic tool that assesses how effectively an institution supports both the structural and personal aspects of the transfer student experience. The TEI is based on a dual-domain framework of challenges focusing on schools and students that were uncovered through Reddit research and survey validation.

Figure 6: Future Directions

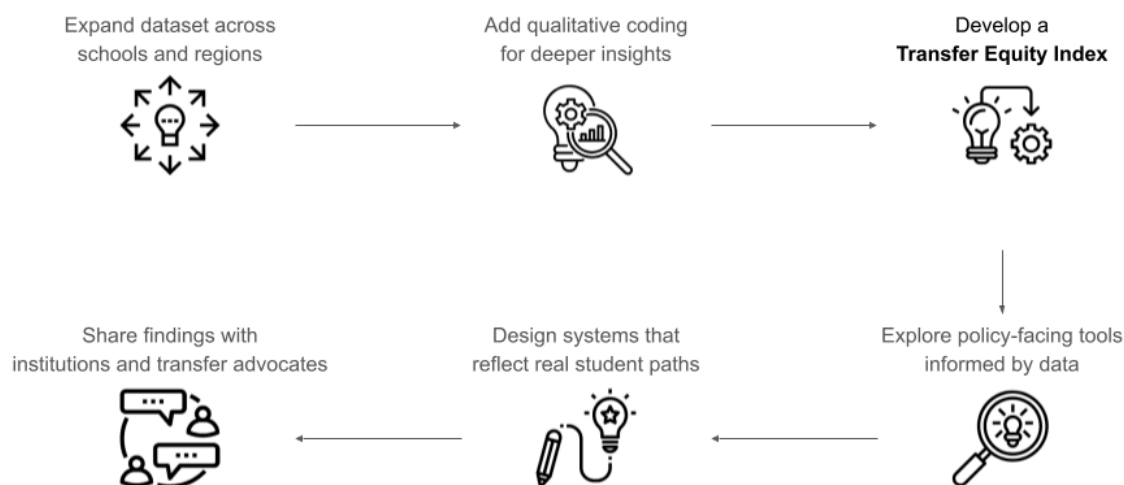


Figure 6 outlines potential future directions for this research. It begins with expanding the dataset across schools and regions, followed by incorporating qualitative coding to generate deeper insights. These efforts support the development of a Transfer Equity Index, which could inform policy-facing tools. Additional steps include designing systems that reflect actual student transfer paths and sharing findings with institutions and transfer advocates to drive systemic improvements.

Components of the Index

The Transfer Equity Index (TEI) is built on sixteen established topics that are equally divided into school-centered and student-centered difficulties. Each theme contributes a weighted score based on three factors: the frequency of problems expressed in Reddit and survey narratives, the sentiment polarity of those expressions (for example, the proportion of negative versus positive posts), and a severity rating manually assigned based on representative posts. The severity of a theme's impact on student transfer success is measured on a scale of 1 (little annoyance) to 5 (transfer-blocking obstacle).

Scoring Methodology

To determine an institution's TEI, researchers or administrators use social media mining or survey tools to gather student input. Concerns are assigned to the 16 TEI categories and rated

based on their observed frequency, emotion, and severity. These dimensions are combined using a weighted algorithm to get a total score ranging from zero to one hundred. The TEI can also be divided into two subindices: School-Centered and Student-Centered, which emphasize different areas of institutional performance.

Example Use Cases

The TEI framework helps to understand why different institutions may struggle. An institution, for example, that receives numerous complaints about complicated credit transfer procedures or unresponsive advising systems may obtain a poor score on the School-Centered Subindex. A college with good academic systems but minimal mental health assistance or poor inclusion of LGBTQ+ or caring students, on the other hand, may underperform on the Student-Centered Subindex. These distinctions assist institutions in understanding not just the difficulties that students experience, but also where their efforts to promote fairness require improvement.

Purpose and Application

The Transfer Equity Index serves a number of objectives. It serves as a diagnostic dashboard for schools and universities, allowing them to discover their most vulnerable areas in servicing transfer students. It also includes a benchmarking tool that will enable establishments to compare their equitable responsiveness to that of their peers. Finally, it provides useful design input for AI-powered advising systems by combining emotional, contextual, and systemic variables that standard academic measures sometimes neglect.

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